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A Survey On Negtive Transfer 2021

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1. Overall Summary

Transfer learning (TL) aims to leverage knowledge from source domains to facilitate learning in a target domain, especially when labeled target data is scarce. However, its effectiveness is not guaranteed, and a phenomenon called Negative Transfer (NT) can occur, where introducing source domain data undesirably degrades target domain performance. This paper addresses the lack of a systematic survey on NT by providing a comprehensive review. It first defines NT and outlines four key factors contributing to it: domain divergence, transfer algorithm, source data quality, and target data quality. The survey then categorizes and analyzes approximately fifty representative approaches for handling NT into four main strategies: secure transfer (with theoretical guarantees), domain similarity estimation (to decide when and how to transfer), distant transfer (for low-similarity domains), and NT mitigation (for medium-similarity domains, focusing on data, model, and target prediction enhancements). The paper also discusses NT in related fields like multi-task learning, lifelong learning, and adversarial attacks. Finally, it offers practical guidelines for coping with NT and suggests future research directions, emphasizing the need for secure transfer in various TL paradigms and NT mitigation for regression problems.

2. Main Idea

The central concept of this paper is to provide the first systematic and comprehensive survey on Negative Transfer (NT) in Transfer Learning (TL). It defines NT, identifies its contributing factors, categorizes and reviews about fifty representative approaches to overcome or mitigate NT (including secure transfer, domain similarity estimation, distant transfer, and NT mitigation strategies), and discusses NT in related fields, offering guidelines for reliable TL.

3. Problem Being Solved

This paper solves the research gap of a missing systematic and comprehensive survey on Negative Transfer (NT) in Transfer Learning (TL). Previous TL research often focused on "what to transfer" and "how to transfer," but largely overlooked the critical problem of "when to transfer" to avoid NT. The paper aims to fill this gap by systematically defining NT, identifying its contributing factors, and reviewing various algorithms and strategies designed to prevent or mitigate it, which was previously scattered across the literature.

4. Assumptions

- The basic assumptions for Transfer Learning (TL) to be effective are: 1) the learning tasks in the source and target domains are related/similar; 2) the source and target domain data distributions are not too different; and 3) a suitable model can be applied to both domains. Violations of these assumptions are key factors leading to Negative Transfer (NT).
- Many NT mitigation approaches assume that the source domain has some similarity with the target domain, which can be estimated.
- The survey primarily considers homogeneous and closed-set classification problems in TL, assuming source and target tasks are the same, and target feature and label spaces are unchanged during testing.
- Some secure transfer approaches assume specific data distributions (e.g., Gaussian distribution for Bayesian adaptive learning) or that source parameters follow a normal distribution.
- Certain domain similarity estimation methods (e.g., test performance based) assume the availability of some labeled target data or an accessible target model.

5. Limitations

- The effectiveness of Transfer Learning (TL) is not always guaranteed, and Negative Transfer (NT) can occur if basic assumptions (related tasks, similar distributions, suitable model) are violated.
- The Negative Transfer Gap (NTG), used to quantify NT, may not always be computable, especially in unsupervised scenarios due to the lack of labeled target data.
- Many existing secure transfer approaches primarily focus on classification problems, with fewer solutions for regression problems.
- Domain similarity estimation methods like MMD only consider marginal distribution discrepancy, potentially failing when conditional distributions are significantly different.
- A-distance, used for similarity estimation, neglects label space differences and only considers marginal distribution discrepancy, which can affect performance.
- Focusing solely on improving feature transferability might lead to poor discriminability.
- Challenges exist in ensuring positive transfer in complex open environments, including continual data streams, heterogeneous features, private sources, unclear domain boundaries, and unseen/unknown categories.
- The survey explicitly excludes articles that do not explain their methods from the perspective of NT to maintain focus.
- In multi-task learning, conflicting gradients can induce NT (negative interference).
- In lifelong learning, models may suffer from catastrophic forgetting or negative transfer from early tasks.
- Adversarial attacks can lead to poor target model performance and NT.

6. Results and Conclusion

The paper systematically reviewed about fifty representative approaches for coping with Negative Transfer (NT) in Transfer Learning (TL), categorizing them into secure transfer, domain similarity estimation, distant transfer, and NT mitigation. Secure Transfer approaches offer theoretical guarantees to avoid NT, ensuring target domain performance is better than without transfer, with examples including Bayesian adaptive learning and regularized least squares. Domain Similarity Estimation is crucial for reliable TL, utilizing methods like MMD, correlation, KL-divergence, A-distance, LEEP, and TransRate to determine when and how to transfer. For scenarios with very low source-target similarity, Distant Transfer employs instance selection mechanisms and auto-encoder pairs to bridge domains. NT Mitigation, for medium similarity, enhances data transferability (domain/instance/feature-level), model transferability (TransNorm, adversarial robust training), and target prediction (soft/selective/cluster-enhanced pseudo-labeling). The comparison revealed that most current NT literature focuses on NT mitigation and domain similarity estimation, primarily through data transferability enhancement. Secure transfer often relies on model adaptation, and many NT mitigation approaches consider multiple NT factors (domain divergence, transfer algorithm, source/target data quality). The paper concludes by providing guidelines for coping with NT based on domain similarity and outlining future research directions.

7. Real World Impact

This research has practical implications for various real-world applications where data scarcity or distribution shifts are common, enabling more reliable and effective deployment of Transfer Learning (TL) systems. This includes improved medical diagnostics (e.g., coronavirus diagnosis from chest X-rays) and brain-computer interfaces (BCIs) by mitigating individual differences and reducing calibration effort. It also contributes to more robust object recognition, face detection, and satellite image analysis by handling varying resolutions, light conditions, and domain shifts. In robotics, it can lead to enhanced robot tracking systems through guaranteed positive transfer of experience. Furthermore, it supports more robust multilingual models by addressing negative interference and provides a general framework for developing more reliable TL algorithms across diverse applications by systematically addressing the problem of Negative Transfer, leading to models that generalize better in new domains with limited labeled data.

8. Graphs and Figures Analysis

Fig. 1. Illustration of NT: introducing source domain data/knowledge decreases the target domain learning performance.

This figure visually demonstrates the concept of Negative Transfer (NT). It shows two scenarios: "Target without source" and "Negative Transfer." In the "Target without source" scenario, a model trained only on target domain data achieves a test accuracy of 70%. In the "Negative Transfer" scenario, when source domain data/knowledge is introduced during training, the model's test accuracy on the target domain undesirably drops to 60%. This illustrates that transferring knowledge from a source domain can sometimes harm, rather than help, the target domain learning performance.

Fig. 2. Our proposed reliable TL scheme.

This flowchart illustrates the paper's proposed scheme for reliable Transfer Learning (TL) by considering strategies to alleviate or avoid Negative Transfer (NT). It starts with "Source Domain(s)" and "Target Domain." The core decision point is "Domain Similarity Estimation," which categorizes similarity into "Low," "Medium," and "High." If similarity is "Low," the scheme suggests "Distant Transfer" or "No TL." If similarity is "Medium," it suggests "NT Mitigation." If similarity is "High," it suggests "Concatenate & Learn." Independently of similarity, "Secure Transfer" is an option that can overcome NT with theoretical guarantees. This scheme provides a structured approach to choose the appropriate TL strategy based on domain similarity to prevent NT.

TABLE 1 Main notations in this survey.

This table provides a comprehensive list of main notations used throughout the survey. It includes symbols for feature vector (x), label (y), feature space (X), label space (Y), source domain (S), target domain (T), probability distribution ($P()$), expectation ($E()$), distance metric ($d().$), loss function ($l()., L()$), hypothesis (h), classifier (f), TL algorithm (θ), error (risk) (ϵ), feature extractor (g), number of samples (n), number of classes (K), and number of source domains (M), along with their respective descriptions. This table serves as a quick reference for readers to understand the mathematical formulations in the paper.

TABLE 2 Existing secure transfer approaches.

This table categorizes and lists existing secure transfer approaches, which are designed to avoid Negative Transfer (NT) with theoretical guarantees. It is divided into "Classification Transfer" and "Regression Transfer." For each category, it lists specific "APPROACH" names (e.g., Adaptive learning, Performance gain, Output truncation, Regularization, Bayesian optimization) and their corresponding "[REFERENCE]" numbers. This table highlights that secure transfer methods exist for both classification and regression problems, providing a quick overview of the state-of-the-art in this area.

TABLE 3 Approaches for domain similarity estimation.

This table summarizes various approaches used for estimating domain similarity, a critical step in reliable Transfer Learning (TL) to decide when and how to transfer. The approaches are categorized into three main types: "Feature Statistics Based," "Test Performance Based," and "Fine-Tuning Based." For each category, specific "APPROACH" names (e.g., MMD, Correlation, KL divergence, Target performance, Domain classifier, Clustering quality, Entropy) and their corresponding "[REFERENCE]" numbers are provided. This table illustrates the diverse methods available for quantifying how similar source and target domains are.

TABLE 4 Approaches for NT mitigation.

This table outlines different approaches for mitigating Negative Transfer (NT) when source and target domains have some medium similarity. The approaches are categorized into "Data Transferability Enhancement," "Model Transferability Enhancement," and "Target Prediction Enhancement." For each category, specific "APPROACHES" (e.g., Domain level, Instance level, Feature level, TransNorm, Adversarial robust, Soft labeling, Selective labeling, Weighted clustering) and their corresponding "[REFERENCES]" are listed. This table provides a structured overview of the strategies employed to improve the transferability of data, models, or predictions to reduce the negative impact of NT.

TABLE 5 Comparison of approaches for mitigating NT.

This table provides a detailed comparison of various approaches for mitigating Negative Transfer (NT), structured by "Strategy" (Domain Similarity Estimation, Secure Transfer, Distant Transfer, NT Mitigation - Data Transferability Enhancement, NT Mitigation - Model Transferability Enhancement, NT Mitigation - Target Prediction Enhancement). For each "[Reference] Approach," it lists its "TL Category" (e.g., Feature adaptation, Model adaptation, Deep TL, Instance adaptation, Adversarial TL), its "Ability to Overcome NT" (rated as *** for guaranteed, ** for very probable, * for possible), and the "Factors of NT" it addresses (D for domain divergence, A for transfer algorithm, S for source data quality, T for target data quality). This comprehensive table allows for a quick comparison of different methods based on their underlying TL paradigm, their effectiveness in overcoming NT, and the specific factors of NT they target.